DataScience101 - Term Project
Student Performance Analysis 3/10/2026

Contents

Introduction 2

Running the Notebooks 2

Milestone1-preprocessing StudentPerformanceNotebook.ipynb 2

Univariate Analysis 3

Univariate Analysis of Categorical Variables – Bar Charts 3

Univariate Analysis of Numerical Variables - Histograms 4

Overall Score Observations 5

Milestone2-eda StudentPerformanceNotebook_EDA.ipynb 5

Bivariate Analysis 5

Box/Violin plots - Numerical VS Categorical Variables 5

Correlation Matrix 6

Relationships between Categorical Variables 7

Summary 8

Inference for a Correlation LinearRegressionTesting.ipynb 8

Overall Score versus Attendance Percentage 8

Overall Score versus Study Hours 9

Milestone-3 Predictions MachineLearning.ipynb 10

Compute a Correlation and Fit a Regression Line 10

overall_score versus study_hours 10

overall_score versus attendance_percentage 10

Multiple linear regression with 2 quantitative explanatory variables: 11

Logistic Regression – target is Grade “A” 11

“A” Grade vs Study Hours 11

“A” Grade vs Parent Education & Internet Access 12

Milestone4- Conclusion – Project.pdf 14

Predictive/Prescriptive Results 14

Introduction

The data for this project is downloaded from www.kaggle.com. At the time it was a trending data set on the kaggle site. It has good reviews for being clean and usable. It is described:

"Filename: Student_Performance.csv Rows: 15,000 Columns: 16

This file contains individually structured student records, where each row represents a single student along with their demographic profile, educational background, learning habits, and academic performance. The dataset combines behavioral, environmental, and academic factors, making it suitable for a wide range of educational and analytical applications.

The file includes information on:

Demographics: age, gender, school type Family background: parent education level Study-related habits: daily study hours, study method, internet access School engagement: attendance percentage, travel time, participation in extra activities Academic records: marks in Math, Science, and English Final outcomes: overall performance score and assigned grade

All values follow consistent formatting, column naming conventions, and realistic ranges to ensure ease of use. The dataset is clean, balanced, and ready for immediate download and analysis."

During the EDA part of the project, I looked more closely at the description of the dataset on Kaggle. It is synthetically generated.

Running the Notebooks

The notebooks were created using Google Colab. The are stored in “FINAL-Data-Science-Project-Student-Performance-Analysis” GitHub repository. Each notebook has a table of contents.

Milestone1-preprocessing StudentPerformanceNotebook.ipynb

The goal was to identify the tabular structure of the dataframe, eliminate duplicates, manage empty cells and outliers. This is also where Univariate Analysis was performed.

The original dataset contained 25000 Rows

The fields:

student_id # of unique student id’s is 15000
age 14 -19
gender - male, female, other
school_type 'public' 'private'
parent_education 'post graduate' 'graduate' 'high school' 'no formal' 'diploma' 'phd'
study_hours 0.5 – 8.0
attendance_percentage 50 – 100 %
internet_access 'yes' 'no'
travel_time '<15 min' '>60 min' '15-30 min' '30-60 min'

extra_activities 'yes' 'no'
study_method 'notes' 'textbook' 'group study' 'coaching' 'mixed' 'online videos'
math_score high score = 100, low score = 0
science_score high score = 100, low score = 0
english_score high score = 100, low score = 0
overall_score high score = 100, low score = 14.5
final_grade ['e' 'd' 'b' 'f' 'c' 'a'] - I am not sure what an "e" grade is

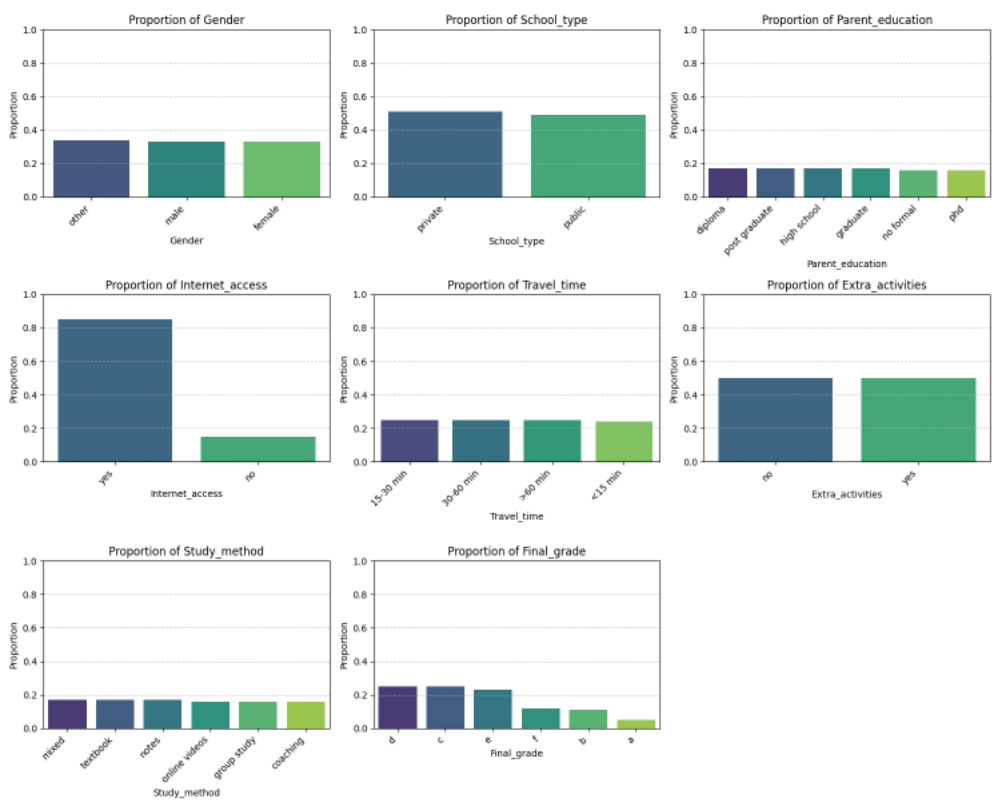
There is no missing data in any column. 10,000 Duplicate Rows – removed

Univariate Analysis

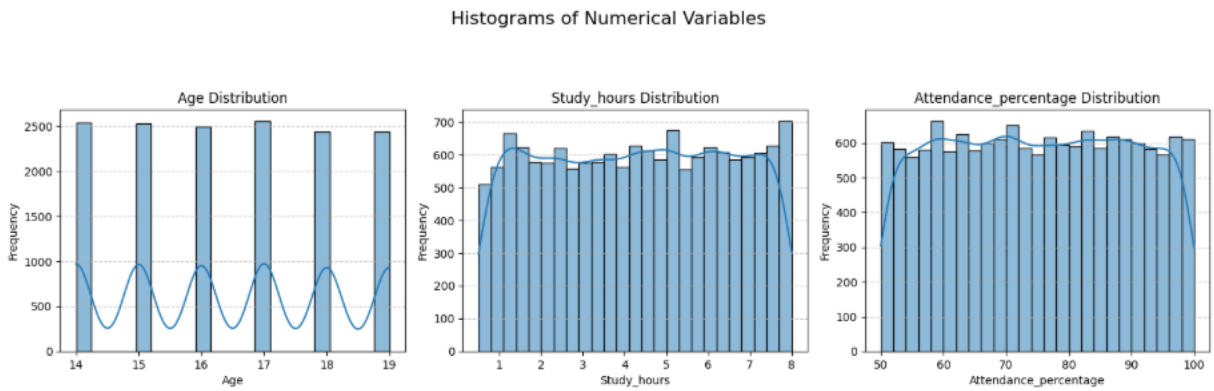
I did Univariate Analysis for both Categorical and Numerical Variables.

Univariate Analysis of Categorical Variables – Bar Charts

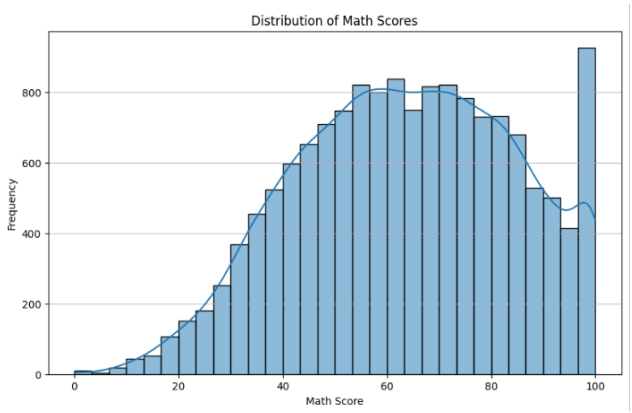
It seems that a stratified sampling method was used. Only internet access and final grades showed any potentially interesting proportional differences.



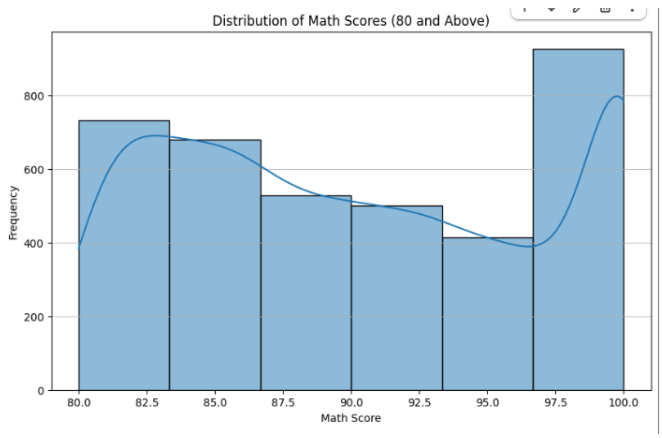
Univariate Analysis of Numerical Variables - Histograms



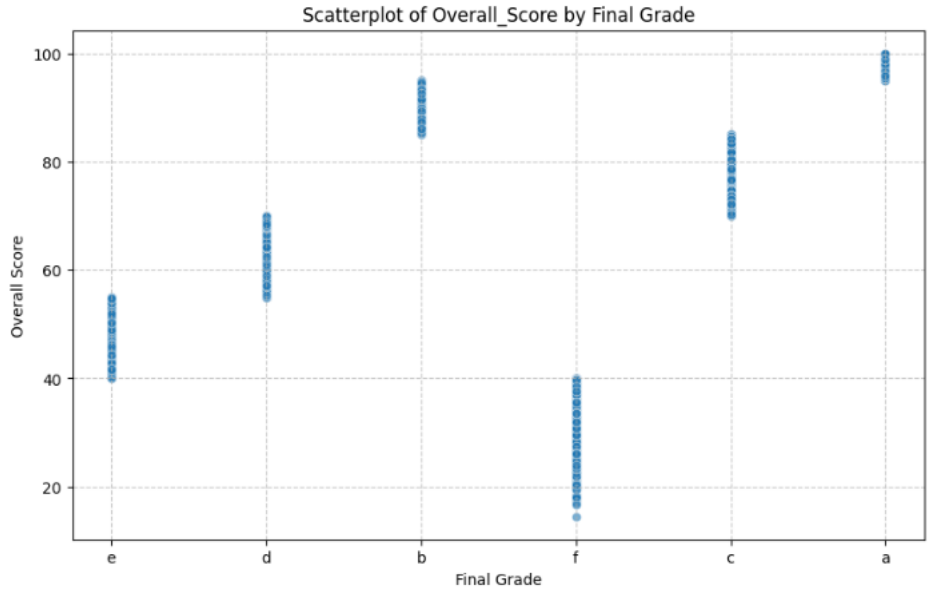
I generated histograms of the overall_scores, math_scores, english_scores, and science_scores. The distribution of the math, English, and science scores all look similar. Here is the one for math.



This is consistent with the high school model where it is expected that multiple students will get a perfect or near perfect score. Close to 1000 students are scoring perfect or near perfect scores.

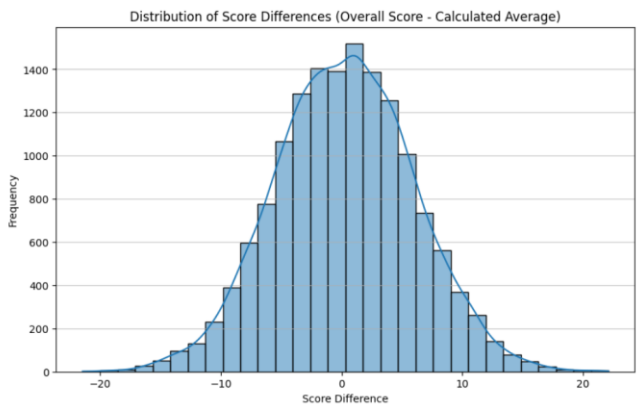


I determined the relationship Between the Overall_Scores and Grades. The grades are related to the overall_score. See below. The grade assignments are consistent with traditional practices, although I am unfamiliar with the “e” grade.



Overall Score Observations

Interestingly, the overall_score is not simply derived from the average math, English, and science scores.



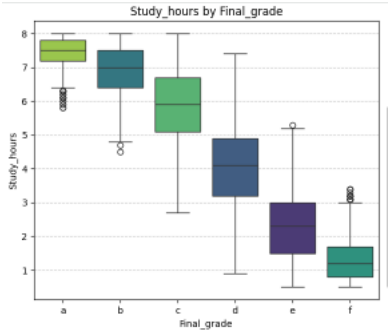
Milestone2-eda StudentPerformanceNotebook_EDA.ipynb

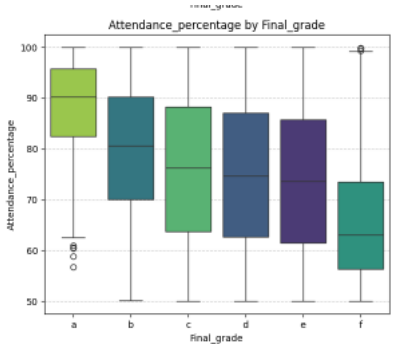
This is where all the Bivariate Analysis was done.

Bivariate Analysis

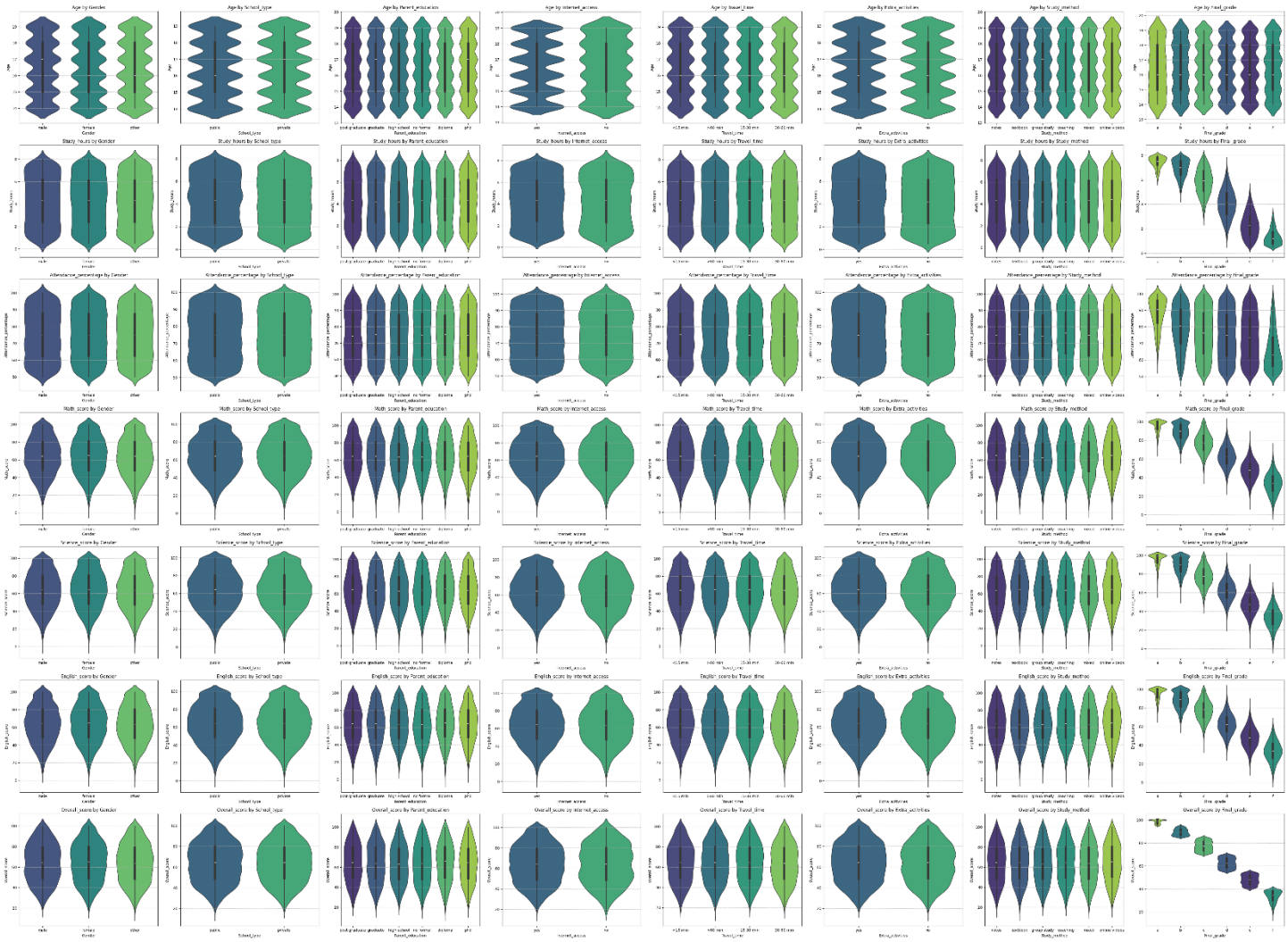
Box/Violin plots - Numerical VS Categorical Variables

I generated a series of BoxPlots (and then later Violin Plots) to Illustrate the Relationship between Numerical and Categorical Variables. Both Study Hours and Attendance Percentage showed a correlation with the final grade. Most BoxPlots (and later ViolinPlots) were uninteresting:



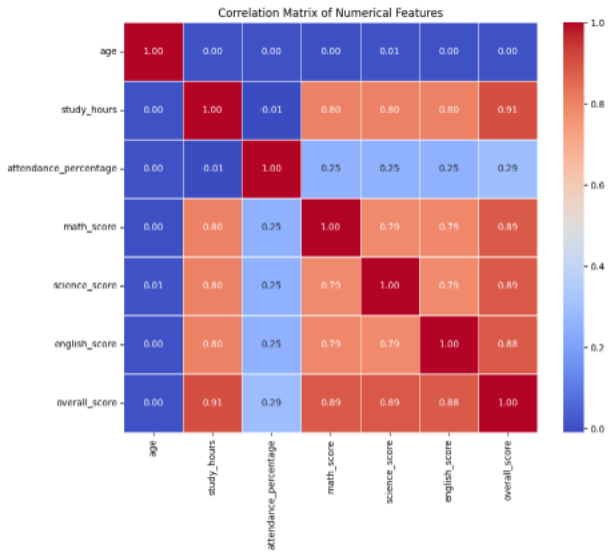


--- Bivariate Analysis: Numerical vs. Categorical Variables (Violin Plots) ---

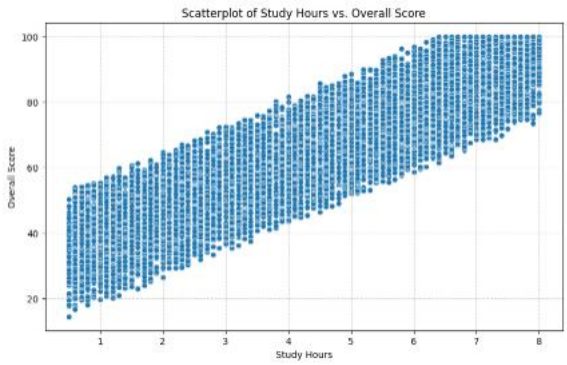


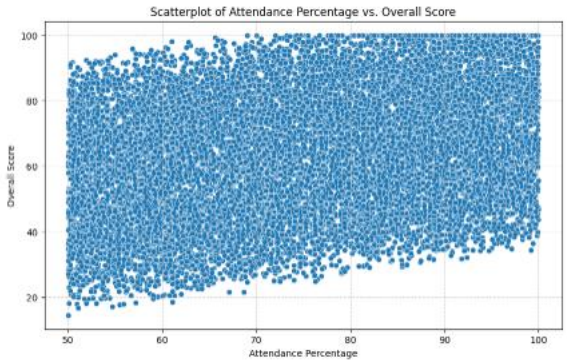
Correlation Matrix

I generated a Correlation Matrix for the Numerical Variables. The student performance was once again linked to study hours and attendance percentage.



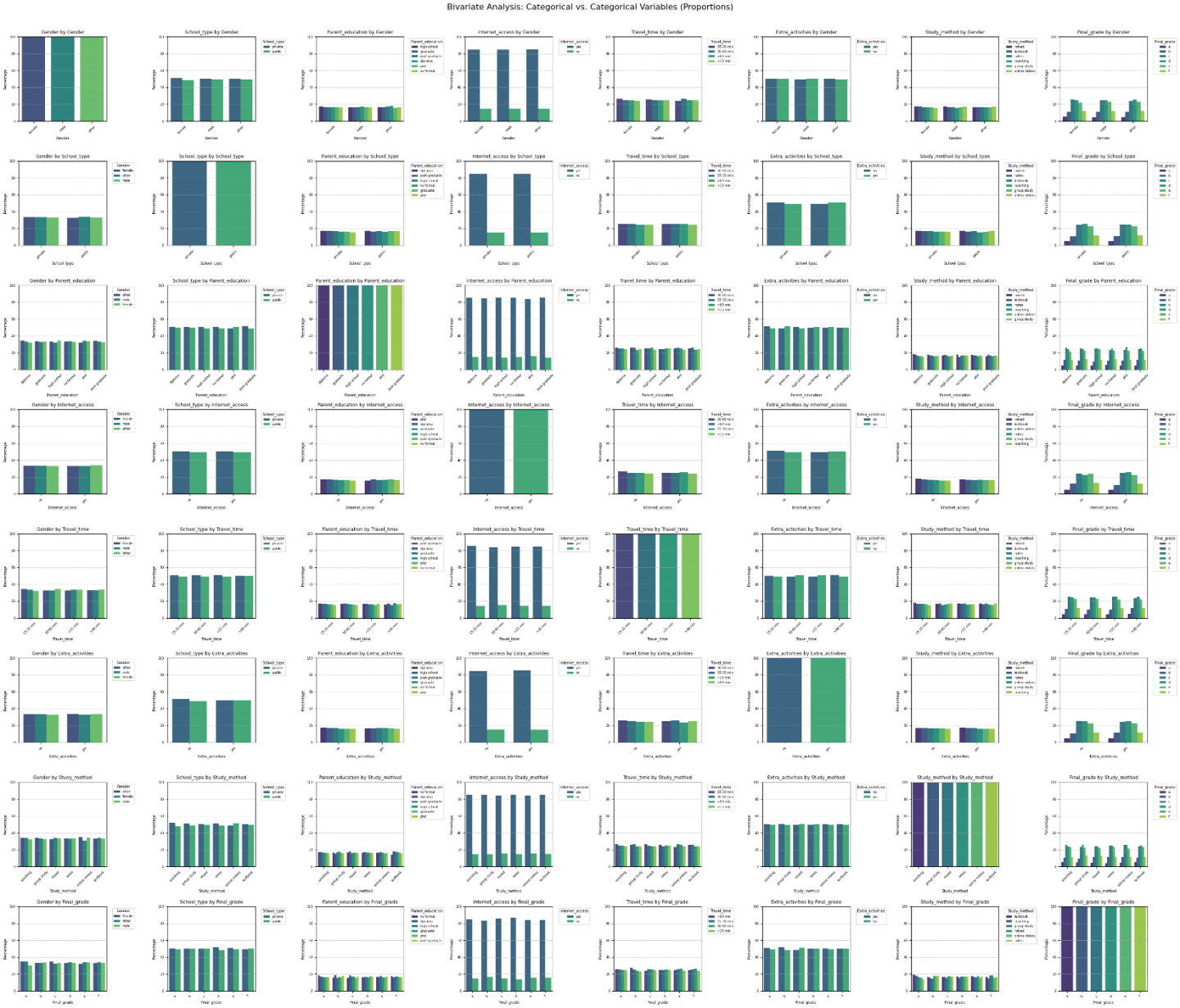
I generated some scatterplots of overall_score versus study_hours and attendance.





Relationships between Categorical Variables

Lastly, I looked for relationship between the Categorical Variables by using grouped bar charts. Nothing seemed particularly interesting. Here are some examples:



Summary

Going forward, I will be focusing on the predictive/prescriptive value of study_hours and attendance_percentage on scores.

Inference for a Correlation LinearRegressionTesting.ipynb

This is where statistical inference was used to evaluate the linear relationship between overall_score and attendance_percentage and then study hours.

Overall Score versus Attendance Percentage

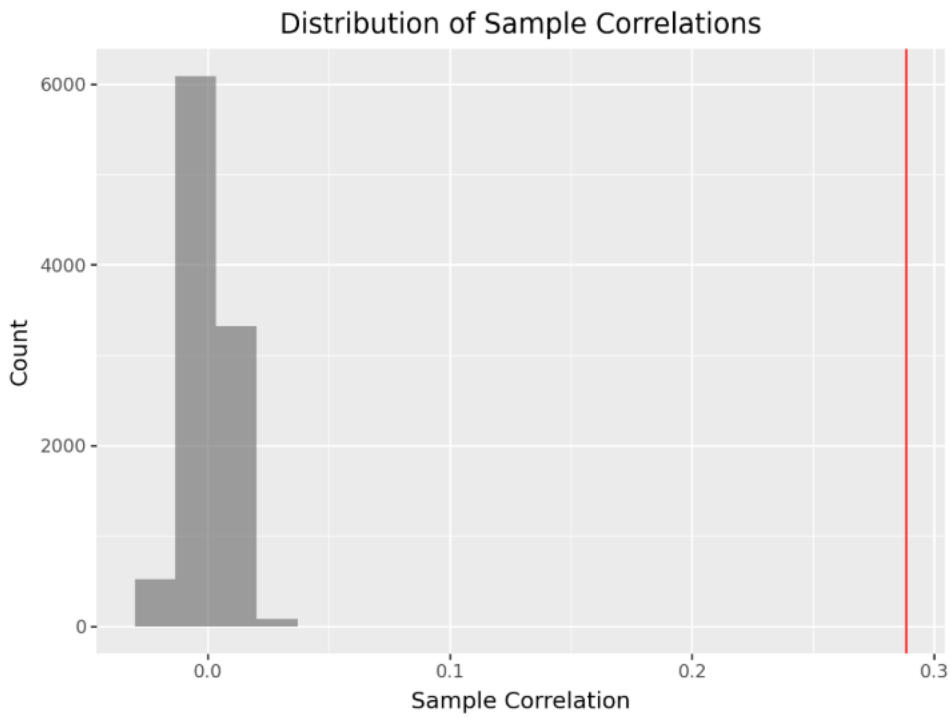
The scatterplot suggests a positive linear relationship. Is there a statistically significant correlation between attendance and score? Is better attendance associated with higher overall scores?

Null Hypothesis: The correlation between overall_score and attendance_percentage is 0.

Alternative Hypothesis: A positive correlation exists between overall_score and attendance_percntage.

The correlation coefficient is 0.28847322776538425

Histogram of 10000 simulated sample Correlations



The proportion of results that are at least as extreme as our sample is essentially 0. With a p-value of 0 there is evidence to reject the Null Hypothesis in favor of the Alternative Hypothesis. There is evidence to conclude that the correlation between attendance percentage and overall score is greater than 0.

Overall Score versus Study Hours

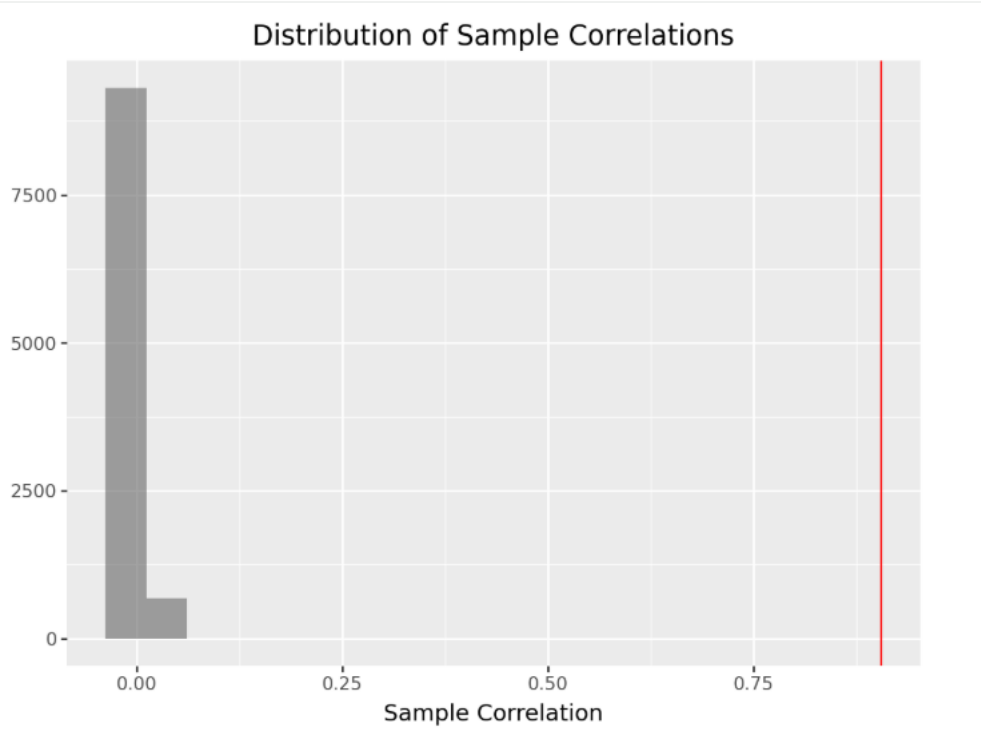
The scatterplot suggested a positive linear relationship. Is there a statistically significant correlation between study_hours and score? Are more study hours associated with higher overall scores?

Null Hypothesis: The correlation between overall_score and study_hours is 0.

Alternative Hypothesis: A positive correlation exists between overall_score and study_hours.

The correlation coeffienct is 0.9058848411126499

Histogram of 10000 simulated sample Correlations



The proportion of results that are at least as extreme as our sample is essentially 0. With a p-value of 0 the is evidence to reject the Null Hypothesis in favor of the Alternative Hypothesis. There is evidence to conclude that the correlation between attendance study_hours and overall score is greater than 0.

Milestone-3 Predictions MachineLearning.ipynb

The original dataset was randomly split (80/20) into a training set and a test set.

Compute a Correlation and Fit a Regression Line

overall_score versus study_hours

Training Data

Regression Line: overall_score = 7.91 * study_hours + 30.31

R-squared value: 0.82 Mean Squared Error (MSE): 64.60

Test Data

Mean Squared Error (MSE) on Test Data for Study Hours model: 65.16

R-squared value on Test Data for Study Hours model: 0.82

overall_score versus attendance_percentage

Training Data

Regression Line: overall_score = 0.38 * attendance_percentage + 35.51

R-squared value: 0.08 Mean Squared Error (MSE): 330.16

Test Data

Mean Squared Error (MSE) on Test Data for Attendance Percentage model: 335.32

R-squared value on Test Data for Attendance Percentage model: 0.09

Multiple linear regression with 2 quantitative explanatory variables:

Training Data:

Regression Equation: overall_score = (17.21 * study_hours) + (5.63 * attendance_percentage) + 64.06

R-squared: 0.91 Mean Squared Error (MSE): 32.72

Test Data:

R-squared: 0.91 Mean Squared Error (MSE): 33.18

Logistic Regression – target is Grade "A"

Confusion Matrix Layout Reminder

For a binary classification, the confusion matrix is typically structured as follows, where rows represent **Actual** classes and columns represent **Predicted** classes:

	Predicted Negative (Class 0)	Predicted Positive (Class 1)
Actual Negative (Class 0)	True Negative (TN)	False Positive (FP)
Actual Positive (Class 1)	False Negative (FN)	True Positive (TP)

"A" Grade vs Study Hours

--- Training Set Performance (A Grade vs Study Hours) ---

Accuracy: 0.92

Classification Report:

	precision	recall	f1-score	support
0	0.95	0.97	0.96	10868
1	0.60	0.48	0.53	1132
accuracy			0.92	12000
macro avg	0.77	0.72	0.74	12000
weighted avg	0.91	0.92	0.92	12000

Confusion Matrix:

[[10499 369]

 [587 545]]

--- Test Set Performance (A Grade vs Study Hours) ---

Accuracy: 0.92

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.97	0.95	2726
1	0.55	0.41	0.47	274
accuracy			0.92	3000
macro avg	0.75	0.69	0.71	3000
weighted avg	0.91	0.92	0.91	3000

Confusion Matrix:

[[2634 92]

 [162 112]]

“A” Grade vs Parent Education & Internet Access

--- Training Set Performance (A Grade vs Parent Education & Internet Access) ---

Accuracy: 0.91

Classification Report:

	precision	recall	f1-score	support
0	0.91	1.00	0.95	10868
1	0.00	0.00	0.00	1132

```
   accuracy                         0.91     12000
  macro avg      0.45      0.50      0.48     12000
weighted avg      0.82      0.91      0.86     12000

Confusion Matrix:
[[10868     0]
 [ 1132     0]]
```

Notice that an “A” grade is never predicted.

```
--- Test Set Performance (A Grade vs Parent Education & Internet Access) ---
Accuracy: 0.91
Classification Report:
              precision    recall  f1-score   support

           0       0.91      1.00      0.95      2726
           1       0.00      0.00      0.00       274

   accuracy                           0.91      3000
  macro avg       0.45      0.50      0.48      3000
weighted avg       0.83      0.91      0.87      3000

Confusion Matrix:
[[2726    0]
 [ 274    0]]
```

Notice that an “A” grade is never predicted.

Milestone4- Conclusion – Project.pdf

This directory contains this document and the PowerPoint Presentation for this project.

Predictive/Prescriptive Results

The most useful predictor of overall_test score is study_hours and attendance_percentage. The analysis suggests that more studying and better attendance improves test scores.

Regression Line Created by our Training Data:

Multiple Linear Regression Equation: overall_score = (17.21 * study_hours) + (5.63 * attendance_percentage) + 64.06

Note that the study_hours and attendance_percentage have been both been scaled to values between 0 and 1. For raw study_hours = 5.5 and raw attendance_percentage = 85.0:

Scaled study_hours = 0.57 Scaled attendance_percentage = 0.69

Predicted overall_score = 77.75

Multiple Linear Regression Model (Training Data):

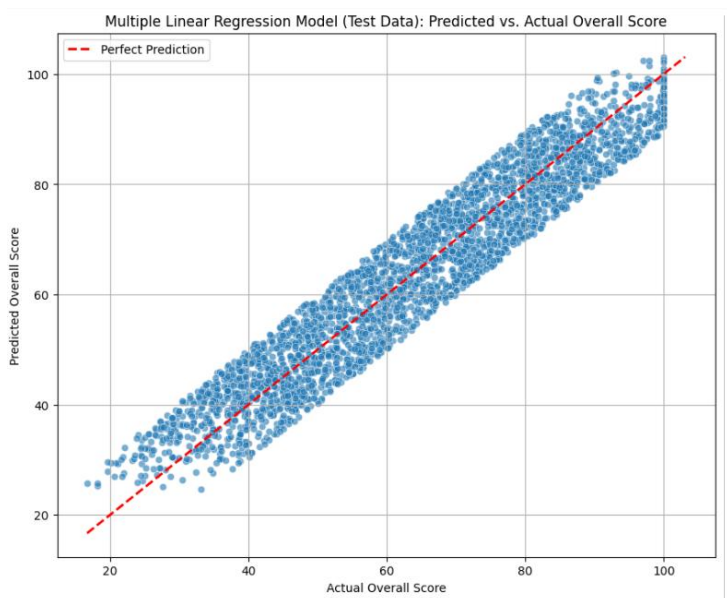
R-squared: 0.91 Mean Squared Error (MSE): 32.72

Multiple Linear Regression Model (Test Data):

R-squared: 0.91 Mean Squared Error (MSE): 33.18

The amount of variation in the overall score that is explained by the model is 91%.

The average distance between our prediction and the actual overall score is 5.7. (square root of MSE)



The results MSE and R squared values are robust and similar for training and test data, suggesting a balanced model.